

GROUP PROJECT



Design and Implementation of a Scalable Cloud - Native System

1. Project Title

Design, Implementation and Evaluation of a Cloud-Based Scalable Application

2. Project Overview

Cloud computing enables organizations to build scalable, resilient, and cost-efficient systems. In this project, student groups will design, deploy, and evaluate a complete cloud-based solution for a real-world problem.

This is a **capstone-style practical + analytical project** requiring architecture design, deployment, storage configuration, virtualization, and implementation of a cloud programming model.

3. Group Structure

- **Group size:** 3–5 students
- Each member must have clearly defined responsibilities
- Peer evaluation will be conducted

4. Project Scenario (Choose ONE)

Each group must select ONE of the following scenarios:

Option A: Cloud-Based E-Learning Platform

Design a scalable e-learning system supporting:

- 10,000+ concurrent users
- Video content delivery
- Online assessments
- Student data storage

Option B: Smart Agriculture Monitoring System

Design a cloud solution that:

- Collects IoT sensor data
- Stores and analyzes crop data
- Provides real-time dashboards
- Sends alerts to farmers

Option C: Cloud-Based Hospital Management System

Design a secure system for:

- Patient records
- Appointment scheduling
- Medical imaging storage
- Role-based access control

Option D: Custom Approved Idea

Groups may propose their own idea (subject to lecturer approval).

5. Project Requirements

◆ PART 1: Cloud Conceptual Analysis – 15 Marks

1. Explain the cloud computing paradigm.
2. Identify:
 - Benefits
 - Risks
 - Challenges
3. Justify why cloud is suitable for your chosen system.
4. Identify compliance or security concerns.

◆ PART 2: Cloud Infrastructure Design – 20 Marks

Design a complete cloud architecture including:

- Compute (VMs, Containers, Kubernetes, etc.)
- Networking (VPC, subnets, load balancers)
- Storage (block, object, file storage)
- Database (SQL or NoSQL)
- Auto-scaling
- Monitoring tools
- Disaster recovery

- ✓ Must include a professional architecture diagram
- ✓ Include explanation of each component
- ✓ Include basic cost estimation

◆ PART 3: Virtualization Implementation – 15 Marks

Demonstrate:

- System virtualization (VM or container deployment)
- Network virtualization (VPC, firewall rules)
- Storage virtualization (virtual disks or managed storage)

Explain:

- Hypervisor type used
- SDN concepts applied
- Software-defined storage concepts

Include screenshots as evidence.

◆ PART 4: Cloud Storage & Data Management – 15 Marks

Implement and demonstrate:

- Cloud object storage (e.g., S3/Azure Blob)
- Database service (RDS, Cloud SQL, MongoDB Atlas, etc.)
- Backup configuration
- Data lifecycle policies
- Access control policies

Explain:

- CAP theorem relevance
- Distributed file system principles
- Data durability mechanisms

◆ PART 5: Cloud Programming Model – 20 Marks

Implement at least ONE of the following:

- Serverless function (AWS Lambda / Azure Function)
- REST API hosted on cloud
- Containerized microservice
- MapReduce-style data processing
- SaaS-style application deployment

Demonstrate:

- Code deployment
- API testing (Postman/screenshots)
- Auto-scaling or event-triggered execution

Explain the programming model used and why it was chosen.

◆ PART 6: Security, Governance & Optimization – 10 Marks

Discuss and demonstrate:

- Identity and Access Management (IAM)
- Encryption (at rest and in transit)
- Firewall/security group configuration
- Cost optimization strategy
- Monitoring and logging

◆ PART 7: Testing & Evaluation – 5 Marks

- Performance testing (basic load test or simulation)
- System limitations
- Proposed improvements

6. Deliverables

Each group must submit:

1. Project Report (4000–6000 words)

Must include:

- Cover page
- Table of contents
- Clearly labeled sections
- Architecture diagrams
- Screenshots of deployment
- Cost estimation table
- References (APA format)

2. Working Cloud Deployment

- System must be live during presentation
- Lecturer must be able to test functionality

3. Source Code Repository

- GitHub/GitLab link
- Clear README documentation

4. Group Presentation (15–20 minutes)

Each member must present.

Presentation should include:

- Architecture walkthrough
- Live demo
- Technical decisions
- Challenges faced

7. Marking Rubric (Total: 100 Marks → 20% CA)

Component	Marks
Cloud Conceptual Understanding (LO1)	15
Infrastructure Design Quality (LO2)	20
Virtualization Implementation (LO3)	15
Cloud Storage & Data Management (LO4)	15
Cloud Programming Model (LO5)	20
Security & Optimization	10
Testing & Evaluation	5
Total	100

8. Technical Requirements

- ✓ Must use a real cloud platform (AWS, Azure, or GCP preferred)
- ✓ Free-tier usage recommended
- ✓ All diagrams must be original
- ✓ Plagiarism < 20% similarity